

GENERAL SPECIFICATION G2.26

Testing and Commissioning

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G2.26 TESTING AND COMMISSIONING

G2.26.1 Overview

- (a) Testing and Commissioning
 - (i) The intent of testing and commissioning is to demonstrate that the Works, including plant/equipment, processes, and systems perform, in all respects, in accordance with the requirements of the Contract; and
 - (ii) During testing and commissioning, training of the Board's personnel in the operation and maintenance of the Works shall also be conducted to facilitate the hand-over of the Works to the Board upon the successful completion of the Works, including any Operations & Maintenance periods specified.
- (b) Testing and commissioning shall comprise:
 - (i) Off-Site testing; and
 - (ii) On-Site testing and commissioning.
- (c) This Specification outlines the general requirements for carrying out the testing and commissioning, the main commissioning and handover stages, the management structures & resources, and key accountabilities for the Contract.

G2.26.2 Stages of Testing and Commissioning

- (a) Testing and commissioning shall be carried out in three (3) stages and will be considered to be completed when each of the following (as relevant) stages is complete; occurs over various stages of the project as outlined below:
 - (i) Off-site Inspection and Testing (Stage 1);
 - (ii) Pre-commissioning Checks (Stage 2); and
 - (iii) Performance Testing (Stage 3).
- Each of the above stages is described in more detail herein.

G2.26.3 Planning for Testing and Commissioning

- (a) Testing and Commissioning Plan

The Contractor shall prepare a Testing and Commissioning (T&C) Plan and submit to the S.O. at least ninety (90) days prior to commencement of the Pre-Commissioning Checks (Stage 2). The T&C Plan shall include as a minimum the following:

 - (i) Scope of testing and commissioning;
 - (ii) Documentation;

(iii) Testing Approach/Methodology:

The methodology in the form of Method Statements shall outline step-by-step procedures for conducting the test and to create the operating conditions, and methods to measure the operational results. Test conditions shall reflect requirements in the Specifications for each item of the Plant. The test plans shall also outline the rationale of each test procedures and how they will demonstrate achievement or failure of the performance requirements.

(iv) Teams – Roles and Responsibilities;

(v) Programme - timelines, and interdependencies of inspection, testing and commissioning activities;

(vi) The sequence of commissioning including the details of the approach for seeding, purging and stabilising the processes along with a risk assessment and associated contingencies and mitigation measures. All interfaces with all contractors where required for commissioning of the entire works shall be detailed;

(vii) Training;

(viii) O&M data;

(ix) Asset Management data; and

(x) A safety response plan including listing of contacts to be notified in case of emergency.

(b) Inspection Test Plans (ITPs)

All commissioning checks shall be administered through ITPs setting out all tests to be undertaken to prove that the Works meets the performance requirements of the Contract. The Contractor shall prepare ITPs for all equipment prior to each stage of commissioning and as a minimum contain asset detail, interlock, safety, electrical and instrumentation checks, and shall also make allowance for the S.O. witness and hold points. All ITPs shall be submitted to the S.O. for review at least thirty (30) days prior to commencing each stage of testing and commissioning.

(c) Pre-Requisites to starting On-Site Testing and Commissioning

No on-Site testing or commissioning activity shall be undertaken by the Contractor until the T&C Plan and ITPs are approved by the S.O..

G2.26.4 Stage 1 – Off-Site Testing and Inspection

(a) Key items of equipment and systems shall be inspected and tested off Site (typically at the manufacturer's premises) to ensure they will function and operate reliably in accordance with the plant design and functional requirements.

(b) The detailed Factory Acceptance Test (FAT) protocol(s) shall be developed by the Contractor to the satisfaction of the S.O.. This includes equipment/systems to be inspected/tested, the testing scope, methodology and acceptance criteria and any requirements for independent approval/witnessing of the tests by personnel with suitable experience and qualifications. The Board or S.O. may also choose to witness such tests. However, in such cases it remains the responsibility of the Contractor to ensure the testing is carried out adequately and to provide suitable personnel for witnessing the tests.

- (c) The Contractor shall offer all items of Plant for inspection and examination by the S.O. at all stages of manufacture and shall include for the FAT of all Plant and equipment as required by the Specification. Inspections and FAT may be (or may not be) witnessed by the Board and/or the S.O..
- (d) The Contract Sum shall include for the Contractor's costs of all off-Site inspections and tests including temporary assembly and subsequent dismantling, labour, materials, instrumentation, chemicals and consumable items, provision of test certificates and certified records.
- (e) No equipment, Plant or materials shall be dispatched from any manufacturer's premises without the written permission of the S.O.. Any equipment, Plant or materials dispatched without such permission shall, if required by the S.O., be returned to the manufacturer's works and shall be tested in accordance with the Specification at the Contractor's expense.
- (f) The off-Site tests shall normally be carried out at the manufacturer's works but if the tests are beyond the resources of the manufacturer, the Contractor shall make arrangements for these to be carried out elsewhere.
- (g) In the event that the S.O. or the Board choose to attend any off-Site inspection or testing, the cost of this will be borne by the Board.
- (h) If during off-Site testing, any equipment, Plant or materials fail to meet the requirements of the Specification, the defects shall be rectified and the equipment, Plant or materials in question shall be offered for re-inspection and witness testing by the S.O. or other inspectors appointed by the Board. All resulting additional expenses incurred by the Board or the S.O. for such re-inspections will be substantiated by the Board and charged to the Contractor through the Contract.
- (i) The Contractor shall carry out all applicable tests required by the Specification and as required by the appropriate reference standards, together with such other tests as are necessary, in the opinion of the S.O. to demonstrate that the equipment, Plant or materials comply with the Specification.
- (j) When the Contractor is ready to carry out any such off-Site test, he shall submit to the S.O. detailed proposals for testing. The proposals shall give as a minimum:
 - (i) A detailed programme for the testing;
 - (ii) Values of test parameters;
 - (iii) Reference to Standards, any other standards and manufacturer's literature;
 - (iv) Proposed format for the test sheets;
 - (v) Certificate of calibration of test instruments; and
 - (vi) Request for S.O.'s approval to proceed with FAT.

- (k) The FAT shall not commence until the Contractor's proposals and test sheets have been approved by the S.O.. After receiving approval, the Contractor shall notify the S.O. of the place and the time for testing, such time being not less than fifteen (15) working days later than the date of the notice for testing in Singapore. For testing outside of Singapore, the Contractor shall notify the S.O. of his intention to carry out the tests at least sixty (60) days prior to the date of testing commencement.
- (l) The Contractor shall carry out every off-Site test at the time and place so notified. If the S.O. does not attend any such test, then the Contractor shall carry out the test in the absence of the S.O. and the certified copies of the test results shall be deemed to be a correct record thereof.
- (m) No inspection or approval for delivery by the S.O. of the equipment, Plant or materials covered by the Contract shall release the Contractor from any of his obligations under the Contract. When Plant and equipment arrives at Site, depending on condition, the S.O. reserves the right to require the Contractor to remove defective Plant, equipment or materials from Site for repair or replacement. The cost of this shall be borne by the Contractor.
- (n) Where items of equipment are of identical size and duty, the S.O. may elect not to witness all tests. The Contractor shall ensure that the performance tests on all items are carried out prior to offering for witnessed testing.
- (o) Calibration of Instruments
 - (i) All instruments used for testing purposes shall have been calibrated by an independent accredited testing authority and shall have a valid calibration certificate. The calibration validity period shall not be longer than twelve (12) months;
 - (ii) Instruments used for tests at site shall have been calibrated not more than three (3) months before the start of testing;
 - (iii) The Contractor shall provide the S.O. with calibration certificates and correction graphs etc. at the time of testing; and
 - (iv) Where reason exists, test instruments shall be re-calibrated when requested by the S.O..
- (p) Test Procedures
 - (i) The procedure and requirements for all tests off-Site on all equipment, Plant or materials shall be in accordance with the requirements of the Specification, and shall in addition be in accordance with the requirements of the reference standards applicable to the items being tested;
 - (ii) The equipment shall be subjected to a thorough visual inspection to confirm that;
 - 1) All components/systems comply with the relevant specifications;
 - 2) All factory assembled components/systems are correctly assembled/ installed in accordance with the relevant Process & Instrumentation Diagram (P&ID) and General Arrangement (GA) drawings/models;
 - 3) All factory assembled pipework joints shall be properly tightened; and

- 4) All warning signs and safety labels shall be fitted in accordance with the relevant GA drawings.
- (iii) Prior to, during or after any test, the Contractor shall adhere to all required safety protocols and standards; including but not limited to method statements and risk assessments.
- (q) **Guaranteed Performance of Plant and Equipment**
 - (i) The figures stated in the Contract for performance and efficiency shall be guaranteed by the Contractor in respect of the equipment offered at the duties specified, and these will be binding and may not be varied except with the written approval of the S.O.. No tolerances are permitted on these figures beyond those envisaged by the specified test standards. Testing to the relevant standards to prove guarantees given will be required for all equipment, Plant and materials and, where applicable, the completed Works. The performance and efficiency shall be guaranteed by the Contractor for the whole duration of the Contract period, including any extension and Defects Liability Period. This shall also include the calibration of instruments.
- (r) **Test Records**
 - (i) For each and every test which the Contractor is required to carry out pursuant to the Contract, the Contractor shall produce a written record in a form approved by the S.O. certifying that the test has been carried out. The Contractor's representative who carries out such tests shall sign the test record. Tests which are witnessed by the S.O.'s representative will be certified by the S.O.'s representative upon the same record provided by the Contractor. In particular, the following must be satisfied:
 - 1) All test results shall be made available to the S.O. within a period of ten (10) working days or unless otherwise specified in the Particular Specification;
 - 2) On completion of testing, all results shall be collated and presented as a single electronic copy and three (3) certified hard copies of the test results with all supporting calculations provided;
 - 3) On each test certificate, sufficient information including the Contract number, type of test, title, all test results, calculations, performance graphs and curves to enable the S.O. to check for compliance with the Specification;
 - 4) Copies of all test certificates shall be included in the operating and maintenance manual; and
 - 5) If any tests are failed, then the report shall identify the cause of failure and detail any remedial actions which have been or will be taken to achieve compliance with the requirements of this Specification.

- (s) Notification
 - (i) Following any inspection or testing of all equipment, Plant or materials, the S.O. shall, within a reasonable time, notify the Contractor in writing either that the item has passed the tests, or that the item on any part thereof is defective, or not in compliance with the Specification or has not met guaranteed performance or efficiency requirements and is rejected, stating the grounds on which the decision is based.

G2.26.5 Stages 2 and 3 – On-Site Testing

On-Site testing shall comprise:

- (a) Pre-commissioning tests – dry (Stage 2a);
- (b) Pre-commissioning tests – wet (Stage 2b);
- (c) Process commissioning – Stage 3a; and
- (d) Plant Performance Testing – Stage 3b.

G2.26.6 Construction to Commissioning Handover

- (a) Before any commissioning activity can commence on-Site, construction in the designated area to be commissioned must be completed to the degree where it is safe for testing to proceed as agreed with the S.O..
- (b) It is the responsibility of the Contractor to coordinate the handover of areas from construction to commissioning.
- (c) All protocols specified in the relevant sections of the T&C Plan for delineating the commissioning area from the construction area shall be adhered to.

G2.26.7 Stage 2a – Pre-Commissioning Testing – Dry

- (a) Inspection and Testing
 - (i) Inspection and Testing involves carrying out all the inspections and tests necessary to confirm the acceptability of the complete installation and its suitability to be operated in a safe controlled manner and meeting the specified design and performance criteria.
 - (ii) Each inspection and respective test shall be clearly identified in the ITPs and be fully documented. The Contractor shall not proceed to the next stage of testing and commissioning until all dry testing has been completed.
- (b) Minimum requirements for Stage 2a
 - (i) Dry testing of equipment involves the checking of equipment, components and ancillaries to establish that they have been supplied and are completely and correctly installed, tested and ready for operation as specified, and in accordance with the manufacturers' recommendations.
 - (ii) Testing shall include, but not be limited to:
 - 1) Hydrostatic tests for tanks, pipework and valves;
 - 2) Operational test for all valves which shall include torque switch setting and functioning test of each electric valve actuator;

- 3) Electrical tests to demonstrate compliance of the electrical installation with specified and statutory requirements. These include checking connection and wiring and drive status conditions of control system. Electrical termination checks of all I/O's, cables etc.;
 - 4) Run tests for all rotating equipment to the extent permitted by the system of which they may form a part – subject to the agreement of the S.O.;
 - 5) Functional testing of all protective devices and setting of protection relays and timers. Including testing of pressure and flow switches and level sensors;
 - 6) Checking of equipment safety devices and interlocks, including necessary safety training or briefing prior to commencement of testing to allow testing to progress safely and effectively; and
 - 7) Individual tests on items of Plant or equipment making up a complete or sub-system. These tests shall include, but not be limited to:
 - Setting and fine-tuning of the Programmable Logic Controller (PLC) and control loops;
 - Setting of protection relays and timers;
 - Checking of instrument readings, both local and on the control system;
 - Testing of pressure and flow switches, and level sensors;
 - Checking of drive status conditions;
 - Checking of control logic and functionality;
 - Checking of equipment start-up and shutdowns (including emergency);
 - Checking of equipment safety devices and interlocks;
 - Checking of automatic sequencing;
 - Measurement of noise levels to verify suitability for unrestricted operation and compliance with noise requirements; and
 - Recording of all settings, timers, etc., both fixed and operator adjustable.
- (iii) The Contactor shall provide all consumables required for carrying out all tests required. The costs for provision of such consumable shall be deemed included in the Contract Sum.
- (iv) The Contractor shall rectify any deficiencies in the equipment design, installation, and performance including operation safety identified during this stage of testing. Any modification or adjustments shall be approved by the S.O. and fully re-tested, recorded in the Testing and Commissioning Report where appropriate.

G2.26.8 Stage 2b – Pre-Commissioning – Wet

- (a) For each complete system or sub-system, after successful completion of Stage 2a – the Contractor shall carry out Stage 2b - wet tests.

- (b) Not all equipment or controls may be able to be tested during Stage 2a, and where this is the case, the Test Plan (Wet Tests) shall clearly state so and moving to Stage 2b shall be at the discretion of the S.O..
- (c) As far as practical, NEWater or plant water shall be used for wet tests.
- (d) The Contractor shall carry out all the necessary adjustments until the components are ready and suitable for starting and running under all operating conditions. Tests shall be carried out to verify that the components will meet operational requirements of the Contract under conditions replicating the operational range as much as possible. The Wet tests shall include any specific tests to simulate the conditions necessary for the proper operation of all Components including stoppages due to simulated power failure.
- (e) During Stage 2b, the Contractor shall test the following as a minimum for each component of each system or sub-system:
 - (i) Process timing sequences will be evaluated and adjusted to suit the dynamic requirements, where possible;
 - (ii) Where appropriate, the equipment will be operated in AUTO and be adjusted via the SCADA system; and
 - (iii) Automatic duty changeover where appropriate shall demonstrate the following;
 - 1) Response to imposed disruptions in operating points and requirements across the full design operating range;
 - 2) Current draw and operational performance of all mechanical equipment;
 - 3) Operation under relevant control modes; and
 - 4) All alarms and control functions are operable (electrical simulation not acceptable).
- (f) Wet Testing shall include the following tasks as a minimum:
 - (i) Adjustment of equipment and control settings;
 - (ii) All instruments have been calibrated and test/calibration sheets have been completed;
 - (iii) Testing of control valves for the full range of flows. The tests will include as a minimum increasing step tests and small and medium step tests. The purpose of these tests will be to assess the compliance of the control valve with the specification and to assess the stability and performance of the valve over(s) the full range of flows;
 - (iv) Site Acceptance Testing (SAT) of the plant control system software;
 - (v) Testing of continuous operation;
 - (vi) Modifications and testing as required;
 - (vii) The operation of mechanical, electrical and control systems under process conditions that represent the anticipated operating conditions;

- (viii) Checking of electrical integrity, including electrical tests for insulation, earth leakage, resistance to high voltage; and
- (ix) Isolation of plant equipment for safe shut-down and maintenance procedures including as a minimum HV station and unit that supplies fire protection systems.
- (g) After successful completion of the Wet Tests, the Contractor shall submit a pre - commissioning report to S.O. for approval with details all tests carried out during Stage 2b. The Contractor shall carry out equipment testing for a period sufficient to provide confidence to the Contractor and the S.O. that the Works is ready to receive used water (Stage 3a).

G2.26.9 Stage 3a – Process Commissioning

- (a) As a prelude to commence this stage of testing, the Pre-commissioning Tests (both Dry Test and Wet Test) ITPs shall all be completed and submitted for S.O.'s approval.
- (b) After successful completion of Stage 2, the Contractor shall seed, purge, tune and stabilise the systems. After stabilising, the plant shall operate uninterrupted and shall be tested to demonstrate the achieving of the design intent for a period sufficient to provide confidence to the Contractor and the S.O. that the Works is ready to undergo Performance Tests (Stage 3b).
- (c) The initial period of Stage 3a will be to commission the parts, and aspects, of the systems that could not be commissioned on clean water. This will require the introduction of seed and the build-up of the biomass for the processes. The start of Stage 3a testing shall commence when the biomass is at the design operational level.
- (d) Process Commissioning is the operation of all equipment systems, control systems and ancillary systems, in conditions that are representative of the anticipated operating conditions. During this time, actual inputs and outputs will be measured to verify that the equipment meets its specified level of performance. Performance of the equipment and ancillary systems, shall be fully documented during this stage e.g. noise, vibration, motor/pump/blower speed, all protection and safety features, alarms and shutdowns, water quality analysers/controllers/monitors, automatic and manual operation. Monitoring and Control Systems, Supervisory Control and Data Acquisition (SCADA) systems and others deemed necessary to complete this stage.
- (e) Process Commissioning shall include commissioning on a unit by unit basis, and then the entire Plant operating. The commissioning could include significant sampling, analysing, testing, monitoring, and recording of data.
- (f) All electrical systems, control systems, subsystems and individual equipment items shall be tested and commissioned to prove the operating capacity of the complete plant. The set points for adjustable devices e.g. sensing relays, shall be recorded on test results and also marked with adjustment points and 'sealed' to prevent movement and unauthorised tampering. Calibrated instruments shall be used for all tests other than logical functions involving only ON and OFF states.

- (g) Results of all tests and the instruments used to conduct the tests shall be recorded and a copy shall be provided within the Operations and Maintenance Manuals. Where initial tests are unsuccessful, the results of these tests together with the corrective measures taken shall also be provided. All test results shall be signed, dated by the Contractor and manufacturer witness by S.O. as where is applicable and submitted to the S.O..
- (h) All proposed commissioning schedules and protocols shall be submitted together with shop drawings, with the Programme. Protocols shall be fully detailed and include at least, schedules in tabulated format covering:
 - (i) Proposed calibration instruments to be used and the tests for which each shall be used;
 - (ii) Functioning of all switchgear to conform correct and uninhibited operations over all stages;
 - (iii) Inspection tests for all meters with 0%, 25%, 50%, 75%, and 100% of full scale readings;
 - (iv) Confirmation of all derived metered values such as hours run, kWh, power factor, kVA etc.;
 - (v) Set points for all adjustable devices such as sensing relays, timers and the like;
 - (vi) Function of all control circuits with all inputs for simulation listed detailing the specified response and with space for recording the actual response;
 - (vii) Function of all monitoring points provided either for immediate or future connection to internal or external systems;
 - (viii) Other specified functional requirements; and
 - (ix) Space on each page for signature and date of witness tests.
- (i) All status, alarm switches, monitoring and control functions shall also be checked. Each device on the loop/logic diagrams shall be signed off by the Contractor's Commissioning Manager after satisfactory completion of the tests.
- (j) The Contractor shall submit signed test results in an approved format confirming the completion of these tests, prior to arranging witness testing by the S.O..
- (k) The Contractor shall provide a completed Testing and Commissioning (T&C) Report prior to commencement of the Plant Performance Test. Test sheets shall constitute the major section of the Report and hence shall commence from the start of the Inspection and Testing stage. A separate report or section shall be provided for each stage of the Testing and Commissioning.
- (l) The report shall follow each of the commissioning stages for each process unit as well as the entire system, i.e. a comprehensive account of findings, results, incidents and activities completed and checked including:
 - (i) All instrumentation and control settings (including P&ID loop control parameters);
 - (ii) Any measurements, checks and settings which may be required by operating and maintenance personnel;

- (iii) Results of any testing and inspection; and
- (iv) Modifications made to any automatic sequences and controls.
- (m) Comments should be included on any observed deficiencies in the plant equipment design and performance and full details of all corrective actions taken shall be recorded.
- (n) Any modifications to circuitry and instrumentation carried out during the commissioning period shall be carefully recorded in detail and provided in the report and recorded on the drawing.
- (o) For the duration of the commissioning period, the contractor shall maintain a daily log of all plant operating data, flows and conditions and record the performance of each item of equipment being tested and/or monitored. All settings and set points shall be listed along with the results of all samples tested.
- (p) The table of results shall as minimum show the performance criteria or duty requirements to be met and the actual performance achieved.
- (q) The S.O. will review the T&C Report and advise the Contractor in writing as to when Stage 3b Plant Performance Test can start. The Contractor shall not proceed to the Plant Performance Test until all Equipment Commissioning operations and performance verifications have been completed and the completed ITP, Commissioning completion report and related documentation submitted and accepted by the S.O..

G2.26.10 Stage 3b – Plant Performance Testing

- (a) The duration of the Plant Performance Test together with the operating regimes and other aspects are described in Particular Specification P10 Testing and Commissioning. The Plant Performance Test will be considered complete when the performance and reliability of the Plant has been clearly demonstrated for the specified duration of the Plant Performance Test. The approach to and the objectives of the Plant Performance Tests are described herein.
- (b) Acceptance Criteria: The acceptance criteria for Plant Performance Tests consists of two (2) components:
 - (i) Performance criteria: As per the Contract specification performance requirement; and
 - (ii) Reliability criteria, concerning overall operation. Refer to relevant Particular Specifications.
- (c) During the Plant Performance Tests, the system shall be operated by the Contractor in accordance with the relevant process control narrative, and other process guideline documents. Collected data will be compared to the performance requirements, with validation of water quality by an accredited laboratory.

(d) Plant Performance Test Plan

Before commencement of performance testing, the Contractor shall submit a Plant Performance Test Plan, which shall describe how the Contractor will carry out the performance testing of the entire system. The proposal shall detail all requirements including the following:

- (i) Commissioning day-to-day program for the performance testing period;
- (ii) Acceptance criteria with performance indicators;
- (iii) Sampling plan and procedures, and management of sampling testing program; and
- (iv) Specific test procedures:
 - 1) Inspections as necessary in the course of the testing, monitoring means, and measurements of key parameters relating to the system's testing including log sheets;
 - 2) Personnel and resources;
 - 3) Communication protocol with emergency contacts;
 - 4) Hazard identification and risk assessment;
 - 5) Safe work method statements;
 - 6) Revised construction management plan; and
 - 7) Emergency/contingency plans.

(e) Plant Performance Test

During the Plant Performance Test, daily review of system performance will be monitored, and the findings shall be tabled by the Contractor in daily meetings in the form of a daily report to the S.O., which shall include information of the previous day's performance testing such as:

- (i) Key operating parameters and any impact on the performance;
- (ii) A log of alarms and any equipment failures;
- (iii) Records of maintenance carried out on assets (if any);
- (iv) Sampling records and results; and
- (v) Compliance with performance and process guarantees.

(f) Based on the information provided by the Contractor in daily meetings, the S.O. will decide if the previous test day is to be considered to have passed or failed.

(g) Failed Days and Extension of Plant Performance Test Period

- (i) Failure to meet any of the required criteria on any day of the Plant Performance Test period, will mean that day is a "failed day";
- (ii) If a "failed day" occurs, the Plant Performance Test period shall be extended by proportionally to compensate;

- (iii) If the number of “failed days” exceeds more than one (1) of the number of days allocated to the testing at a particular plant testing condition, then the testing at that production rate shall be restarted from Day 1. For example, if ten (10) days is allocated to testing at the maximum plant production rate, the ten (10) day test period will restart from day 1 if more than one (1) “failed day” occurs;
 - (iv) Progress through the Plant Performance Test Period will be progressive at the production rates and operating scenarios described in Particular Specification, P10 and progress will depend on proving performance at each stage. The S.O. will determine when progress to the next stage can occur; and
 - (v) All costs related to retesting due to consequence of a failed day shall be borne by the Contractor.
- (h) Plant Performance Test Report
- After successful completion of the Plant Performance Test, the testing data shall be documented in the form of a report which details developments and outcome of the performance testing. The report shall include details of test results, and where applicable, descriptions of modifications to the system and experienced gained from its operation. The report will include:
- (i) The details of the reliability and performance criteria conformance;
 - (ii) Sampling test results;
 - (iii) Laboratory sampling report;
 - (iv) Testing log and check sheets;
 - (v) Completed ITPs;
 - (vi) Final commissioning set points, if changes carried out during the process proving period;
 - (vii) Plan to amend the handover documentation including ‘as built’ drawings based on any modifications carried out; and
 - (viii) Experience gained from the testing will be shared with the plant operations and maintenance team.

G2.26.11 Testing, Sampling and Analysis Methods

- (a) Sampling
 - (i) During the various stages of testing and commissioning, the Contractor shall collect samples and carry out analyses for the listed parameters in accordance with the Contract sampling requirements.
 - (ii) The methods to be used to analyse the listed parameters shall be as given in Standard Methods for the Examination of Water and Wastewater; 21st Edition, 2005; published by APHA, AWWA and WEF.

- (iii) The Contractor may either use the laboratory at the Works or choose to conduct these tests using laboratories with accreditation for the required tests. Where the Contractor intends to carry out the tests at the Works laboratory, the Contractor shall provide the required skilled laboratory technicians supervised by an experienced chemist to perform such tests at such hours the operation may require. The S.O. reserves the right to reject personnel considered not suitably qualified or experienced.
- (iv) Where an external accredited laboratory is used for testing of samples, the Contractor shall supply, as part of the T&C Plan, a copy of the Quality Manual prepared by the Laboratory that will conduct sampling and analysis for the Contract.
- (v) The Contractor shall complete a chain of custody form for all samples taken, at the time of carrying out the work. Sample containers shall be suitably labelled to identify the time, date etc.
- (vi) Each sample taken shall be divided into three and stored in separate containers; one container shall be used for analysis by the Contractor, in accordance with the schedule. This analysis may be witnessed by the Board's chemist. The second container may be used by the Board for analysis at their discretion. The third container shall be retained as a check sample. The overall sample volume shall be at least 3l – split equally between containers. All check samples shall be stored by the Contractor in a fridge on Site for a period of 48 hr after completion of analysis.
- (vii) The Board may, at its discretion, carry out additional sampling and tests at its own laboratories or accredited laboratories as it deems necessary.
- (viii) The S.O. reserves the right to discard obviously high or low erroneous test results.
- (b) Water and sludge samples - used water, mixed liquor, MBR filtrate, RO permeate, industrial water, NEWater, final effluent, sludge and sludge cake samples.
 - (i) The Contractor shall use refrigerated samplers for sampling at selected sampling points specified in the contract documents. All sampling for chemical analysis shall be flow weighted composite unless otherwise specified.
 - (ii) The automatic samplers shall be programmed to collect maximum volume over the sampling day, i.e., 9.5l.
 - (iii) The samplers are required to function correctly and collect the correct amount of sample in the correct manner throughout the testing period. Failure of the sampling renders the results invalid and is considered a defect requiring the testing to be restarted after the sampling failure is rectified.
 - (iv) Grab sampling consists of either a single discrete sample or individual samples collected over a period of time not exceeding 15 minutes. The grab sample should be representative of the wastewater conditions at the time of sample collection.
 - 1) Manually collect a sample by using the actual sample container which will be used to transport the sample to the laboratory. This eliminates the possibility of contaminating the sample with intermediate collection containers.

- 2) If the wastewater stream cannot be physically reached by the sampling personnel or it is not safe to reach for the sample, an intermediate collection container may be used, from which the sample can be redistributed to other containers. If this is done, however, the container used to collect the sample must be properly cleaned and must be made of a material that meets the requirements of the parameter(s) being investigated.
 - 3) Samples for oil and grease, bacteria, and most volatile compounds must always be collected directly into the sample container. The container should not be overfilled if preservatives are present in the container.
- (v) Composite sampling consists of a series of “grab samples” (aliquots) that are collected over a production or discharge day, and are combined together into a single, composite sample. A composite sample represents the average wastewater characteristics during the compositing period. Various methods for compositing are available and are based on either time or flow proportioning.
- 1) A time composite sample consists of equal volume discrete sample aliquots collected at constant time intervals into one container. A time composite sample can be collected either manually or with an automatic sampler.
 - 2) A flow proportional composite sample can be collected using one of two methods. One method consists of collecting a constant sample volume at varying time intervals proportional to the wastewater flow. For the other method, the sample is collected by varying the volume of each individual aliquot proportional to the flow, while maintaining a constant time interval between the aliquots.
 - 3) At the end of the compositing period, the sample collected should be properly mixed and transferred into the respective containers, followed by immediate preservation, if required.
- (vi) Sampling equipment must be properly cleaned to avoid cross-contamination which could result from prior use.
- (vii) The methods to be used for collection and preservation of samples shall be as given in Standard Methods for the Examination of Water and Wastewater; 21st Edition, 2005; published by APHA, AWWA and WEF.
- (c) Screenings Samples
- (i) There is no standard method for measurement of the BOD content of washed and dewatered screenings. The BOD content in the washed and dewatered screenings shall be measured by the BOD₅ and expressed as a cleanliness factor or concentration as below:

$$\text{Cleanliness Factor} = \text{BOD}_5 \text{ (g/l)} / \text{Density (t/m}^3\text{)} \times \text{Dry Solids Fraction}$$
 - (ii) The Contractor shall guarantee that eight (8) out of ten (10) samples shall have a cleanliness factor of less than fifty (50).

- (d) Grit Samples
 - (i) The moisture content and the organic content of the washed and dewatered grit shall be measured in accordance with requirements in Particular Specification P4 Process Mechanical Works.
- (e) Odour Samples
 - (i) The Contractor shall sample each odour source and the samples shall not be collected within five (5) days of any significant rain event in the catchment of the sewerage system.
- (f) Noise Samples
 - (i) Occupational Noise Measurement

The Contractor shall carry out noise analysis on following equipment as a minimum:

 - 1) Pumps with motors greater than 5 kW (unless submerged);
 - 2) Pneumatic Valves;
 - 3) Aeration blowers and pipework;
 - 4) Air Compressors;
 - 5) Air Conditioners and Exhaust Fans; and
 - 6) Any mechanical equipment with a motor power greater than 15 kW (unless submerged).
 - (ii) Noise Analysis on equipment shall be as follows:
 - 1) An A-weighted, and octave band spectrum analysis (Lin), from 63 Hz to 8000 Hz shall be measured no further than 1 m from the operating unit. If the sound pressure level in any octave band exceeds the sound pressure level in both adjacent octave bands by more than 5 dB (i.e. a pure tone is dominant) then the respective measured overall weighted sound pressure level in dB(A) shall be increased by 5 dBA;
 - 2) The Noise readings shall be taken at the most disadvantaged location(s);
 - 3) The sound pressure levels shall be measured with 'Type 1 precision integrating sound level meter' conforming to the local standards and calibrated within the last two (2) years; and
 - 4) All measurements shall be carried out by trained personnel.
 - (iii) Industrial Noise Measurement

The Noise measurements shall be in accordance with the Singapore standards requirement:

 - 1) Noise measurements shall be at a time when conditions relating to metrology and operation of the works will yield the highest sound level;
 - 2) Noise measurement on property boundaries to the site shall be at the location on that boundary which will record the highest sound level;
 - 3) Noise measurements shall be aborted if wind is more than 18 km/h or if rain occurs; and

- 4) All measurements shall be carried out by trained personnel.
- (g) Vibration Testing
- (i) Vibration testing shall be carried out and results provided by the contractor on all installed equipment. The vibration tests shall be carried out over the full operation range and full operational load and be captured in the ITPs. The vibration tests shall be conducted and evaluated in accordance with relevant standards applicable to newly commissioned assets/equipment. The Contractor shall supply full details of the test procedure, conditions and standards used.
 - (ii) Onsite vibration testing will not be required for submersible pumps and mixers mounted in a wet wells, reservoirs, storages and channels etc. unless pump has remote monitoring points fitted.
- (h) Water Retaining Structures
- (i) The Contractor shall carry out hydrostatic tests to determine if there are any leaks in water retaining structures. Water retaining structures shall include concrete pump wells, concrete or steel/stainless steel tanks and other similar structures. Water retaining structures shall be tested to maximum water height, which will be to the top of the containing walls or to weir overflow level. Unless approved by the Superintendent no backfilling shall occur around the structure until the structure has satisfactorily passed the hydrostatic test to the satisfaction of the S.O.. Any material that may inhibit a full perimeter inspection of the structure during testing shall be removed unless it can be demonstrated that it is necessary for structural or safety reasons.
 - (ii) The hydrostatic testing for concrete structures shall be carried out in accordance with Singapore Standards. Timeframe for filling the tanks for hydrostatic testing shall be the responsibility of the Contractor.
 - (iii) Concrete structures must be filled at a rate of no more than 2 vertical metres per 24-hour period until the maximum hydraulic water level is reached. When full the structures shall be inspected for any sign of leakage. If leakage is detected the tanks shall be emptied as required and repair made and the test repeated until there is no visible leakage. Once visible leakage has been stopped concrete structures shall be left full of water for seven days prior to commencing the leakage test. The leakage test shall be carried out for a period of seven days and net loss of water calculated taking account of evaporation. For concrete structures, if leakage is greater than 0.2% of the maximum operating water depth for that compartment, the structure shall be deemed to have failed the hydrostatic test and the Contractor shall carry out appropriate repairs and retest the tank. For steel structures, there should be no leakage apart for evaporation. The S.O. will advise the tolerance to be allowed for in assessing evaporation. The testing shall be repeated until the structure passes the test.
- (i) Gas Retaining Structures
- (i) The Contractor shall carry out gas-tight tests to determine if there are any leaks in gas retaining structures. Gas retaining structures shall include digesters, ozone reaction tanks and other similar structures.

- (ii) Gas retaining structures shall be tested at two stages:
 - 1) Test 1 - Prior to any interior or external coatings being applied: to identify concrete cracks;
 - 2) Test 2 - After all protective coatings have been applied: to test the structure for gas tightness at the maximum operating pressure.
- (iii) Prior to the test, all embedded pipes and other equipment shall be installed. All openings into the gas area shall be temporarily blanked off and sealed.
- (iv) One of the access panels, or pipes, shall be blanked off and sealed with a temporary metal plate. To this plate four pipes shall be fitted:
 - 1) A pressure relief valve set at the design relief pressure of the structure. This valve can be omitted if there is hydraulic over-pressure protection of the structure;
 - 2) A low pressure manometer;
 - 3) A compressed air supply connection fitted with a non-return valve; and
 - 4) A thermometer or thermocouple to measure the temperature of the internal gas space.
- (v) The structure shall be filled up to the minimum operating water/fluid level, including allowance when operating at the maximum operating pressure. Unless approved by the Superintendent no backfilling shall occur around the gas volume of the structure until the structure has satisfactorily passed the gas-tight tests to the satisfaction of the S.O.. Any material that may inhibit a full surface inspection of the structure during testing shall be removed unless it can be demonstrated that it is necessary for structural or safety reasons.
- (vi) The structure shall be pressurised with compressed air to the maximum operating pressure, with a step at +3 kPa and at a maximum rate of 1 kPa per hour.
- (vii) Test 1
 - 1) The bare concrete of the outer surface of the structure shall be sprayed with a mixture of water and surfactant liquid soap.
 - 2) After the soapy water spray, a visual check of all sprayed surfaces (roof as well as the upper part of the walls corresponding to the internal gas space) shall be undertaken.
 - 3) All zones of significant leakage shall be marked and repaired. Significant is determined by the extent of leakage and shall be subject to the agreement of the S.O.. As a guide, these zones will have a surface area >100 cm² and bubbling thickness >2cm.
 - 4) This test shall be repeated until there are no significant areas of leakage.
- (viii) Test 2
 - 1) The bare concrete of the outer surface of the structure shall be sprayed with a mixture of water and surfactant liquid soap.

- 2) After the soapy water spray, a visual check of all sprayed surfaces (the roof as well as the upper part of the walls corresponding to the internal gas space) shall be undertaken.
 - 3) The pressure in the gas space shall be monitored to establish if the pressure is maintained. Maintaining pressure is defined as a pressure decrease of less than 0.25% in 1h or of less than 5% in 24h, at constant temperature. In the case of temperature variation, the pressure shall be corrected using the equation $P_2 = P_1 \times \frac{T_2}{T_1}$ where the temperatures T_1 and T_2 are expressed in °K.
 - 4) If the pressure is not maintained and/or if any bubbles are detected the structure shall be deemed to have failed the gas-tightness test.
 - 5) Any leakage anomalies shall be pinpointed and be repaired and the test repeated until passed.
- (j) Hydraulic Capacity
- (i) The Contractor shall carry out tests necessary to determine compliance with the Contract for the hydraulic capacity of process units and systems including flow splitting and bypasses, under simulation of the most severe loading conditions. The Contractor shall provide, calibrate and maintain all flow measuring devices, which are necessary for these tests. The flow measuring devices shall be installed in a manner to ensure accurate measurement. Flow splitting between process units and overflow weir settings shall be within +/- 5% of that required by the design inclusive of the accuracy of the flow measuring devices.
 - (ii) If peak instantaneous hydraulic conditions are not achievable during commissioning, the Contractor shall test hydraulic capacity at the maximum flows achievable, as agreed with the S.O. and submit calculations, which predict the performance at the maximum design flows by calibration of the design calculations and/or extrapolation of the results obtained for the flows used in the hydraulic capacity tests to the peak instantaneous hydraulic conditions.
- (k) Pipework
- (i) The pipework between pump(s), blower(s) and the limit of contract shall be tested in accordance with Singapore Standards.
 - (ii) Pipework shall be tested for leakage and defects in the pipe, joints, fittings and valves. The test shall be carried out in sections as soon as practicable after each section has been installed, jointed and cleaned, and after erection and before concreting. The Contractor shall ensure that the pipework is suitably anchored and supported to sustain the pressure in the not-built-in condition.
 - (iii) Pipework and valves through which oil or gas or liquefied gas is to be conveyed shall be thoroughly inspected, then tested with the oil with which they are to be used or an inert gas and checked for leakage. Unless otherwise specified, hydraulic tests using water shall NOT be applied to these types of pipework.
 - (iv) Solvent cement joints shall be cured for at least 24 hours before testing.

- (v) To achieve stable testing conditions, the pipe section to be tested shall be filled slowly with water, ensuring all air is expelled and allowing for absorption. The section shall be kept full of water for 48 hours for cement mortar lined pipes, or 24 hours for other pipes before testing.
 - (vi) During pressure testing of a pipework, each isolation valve shall sustain the full pressure on one side of the valve with no pressure on the other side for at least 15 minutes.
 - (vii) The hydrostatic test pressure shall be maintained for not less than 2 hours, and the timeframe for filling the tanks for hydrostatic testing shall be the responsibility of the Contractor.
 - (viii) During the test, there shall be no visible leakage or sweating at joints nor must any defects in the Contractor's work become apparent. Any such defects shall be repaired by the Contractor at its own cost and the test repeated until the Superintendent is satisfied that the whole work is free from defects.
 - (ix) The minimum hydrostatic test pressures shall be:
 - 1) Pressure Pipelines: 1.5 times the maximum possible working pressure of the pipework;
 - 2) Chemical Pipelines: 1.2 times the maximum possible working pressure of the pipework;
 - 3) Oil or gas or liquefied gas Pipelines: 1.2 times the maximum possible working pressure of the pipework; and
 - 4) Gravity Pipelines: Maximum static pressure possible.
 - (x) Where appropriate, valves shall be pressure tested when pipelines are pressure tested. All valves of nominal size of 100 mm or larger shall be hydrostatically tested at the manufacturer's works and provided with a certification from the Manufacturer that the valve has been successfully tested in accordance with the relevant Standard.
- (l) Pump Testing
- (i) Prior to the commissioning of each pump, it shall be tested by pumping for a short period against a head similar to that expected in the duty. During this initial run, the pump shall be observed for excessive noise, vibration and leaks. When the pump satisfactorily meets the requirements above, the pump shall be run continuously with water for a period of four (4) hours or a shorter duration subject to the approval of the S.O. during which the following shall be measured:
 - 1) Bearing temperature;
 - 2) Power consumption of electric motors; and
 - 3) Seal operation.
 - (ii) The accuracy of metering pump/feeder settings versus flow/feed rates graphs for all metering pumps/feeders shall be verified.
 - (iii) The Contractor shall provide pump test certificates from the pump manufacturer for all pumps of greater than 1 l/s capacity as follows:

- 1) Casing Hydrostatic Test Certified; and
 - 2) Pump Performance Test Certified to the relevant standards.
- (m) Blower Testing
- (i) Prior to the commissioning of each blower, the blowers shall be tested by operating them for a short duration against a discharge pressure similar to that expected under normal operation. During this initial run, the blower shall be observed for excessive noise, vibration and leaks. When the blower satisfactorily meets the requirements above, the blower shall be run continuously for a period of four (4) hours or a shorter duration, subject to the approval of the S.O., during which the following shall be measured and recorded:
 - 1) Ambient temperature, pressure and relative humidity;
 - 2) Inlet/outlet temperatures;
 - 3) Inlet/outlet pressure;
 - 4) Discharge flow;
 - 5) Bearing temperatures;
 - 6) Vibration readings;
 - 7) Amperage draw;
 - 8) Power consumption;
 - 9) Seal operation;
 - 10) Oil temperature; and
 - 11) Oil pressure, etc.
 - (ii) The test shall demonstrate that the blowers will be operated over the entire range of specified pressure and flow while in remote control mode from the SCADA. The test shall be conducted with the basins full of water (such as, plant effluent) at normal operating levels.
 - (iii) Conduct a minimum 2-hour blower run test to demonstrate various operating scenarios. The test shall demonstrate that the blowers are able to sequence on and off as lead and lag systems as required to maintain the setpoint (such as, flow, pressure or DO, etc.) received from SCADA.
 - (iv) Demonstrate the ability of each blower to automatically restart following a plant wide power failure without operator interface.
 - (v) A Field Startup Log shall be submitted on completion of each test which records the blower model number, blower serial number, test date, beginning test time, ending test time, motor power, motor speed, amperage draw, and all of the key operating parameters specified in the Design Criteria. Test results shall be certified by the manufacturer/manufacturer's representative and endorsed by the Contractor and witnessed by the S.O..
- (n) Leak Testing of Ductwork and Adjustment

- (i) Leak testing of the ductwork will be carried out to identify missing gaskets or poor jointing and connections;
- (ii) After all ductwork is assembled, the suction isolation valves for each suction point shall be placed in the closed position. Thoroughly inspect the ductwork in the selected branch for any leakage. If leakage is identified, the defect shall be corrected and the procedure repeated until there is no leakage from the ductwork; and
- (iii) Flow balancing shall be carried out so that adjustment of dampers to their final positions can be carried out to achieve the flow that matches the design flow rates.

G2.26.12 General Requirements for Testing and Commissioning

- (a) Key items of equipment and systems shall be inspected and tested off site. The Contractor shall provide all necessary resources and materials including labour to operate and attend to all testing and commissioning activities during all stages of work until the issuance of the Certificate of Substantial Completion and including attending to defects during the Defects Liability Period (DLP) until the issuance of the Final Completion Certificate for the whole of the Works. The costs for provision of such resources and materials shall be deemed included in the Contract Sum.
- (b) Where required as part of Testing and Commissioning, during the Plant Performance Period, Interim Operating Period and Plant Proving Period, the Contractor shall provide labour on three shifts, twenty-four (24) hours a day, seven (7) days a week when required to operate the Plant to demonstrate performance. The costs for provision of such resources and labour shall be deemed included in the Contract Sum.
- (c) Unless specified otherwise, the Contractor shall provide all necessary materials, chemicals, electricity, fuel, stores, apparatus, instruments and indicators necessary to carry out the tests. Water and electricity used for such tests shall be recorded on suitable instrumentation/meters provided by the Contractor, which shall be read by the S.O. in the presence of the Contractor at appropriate times. Water and electricity consumed as a result of such testing activities shall be paid for by the Contractor at the appropriate commercial rate, to be determined by the S.O.. The S.O. will from time to time deduct such agreed cost of supplying water and electricity to the Contractor from the Contractor's monthly progress claims. The connection points for the tapping of electricity and water shall be confirmed by the S.O..
- (d) Tests and inspections, unless otherwise specified, shall be in accordance with the latest relevant Singapore Standards or in the absence of local standards, recognised international standards shall apply.
- (e) Hold Points and Witness Points
 - (i) A Hold Point is defined as a position in the progress of the work under Contract, beyond which further work shall not proceed without the approval of the S.O.; and

- (ii) A Witness Point is defined as a position in the progress of the work under Contract, where the Contractor must notify the S.O. prior to proceeding and the option for attendance for witnessing of inspection and test may be exercised. If any do not attend, then work may nevertheless proceed, unless otherwise instructed. Witness Points shall apply to verify compliance of the constructed works with the approved design drawings.
- (f) The Contractor shall agree Hold Points and Witness Points with the S.O. based on the Contractor requirements and shall ensure compliance with the intent of the designs and with other specified requirements, and to ensure that critical and/or irreversible activities are not constructed incorrectly.
- (g) The S.O. reserves the right to witness any or all tests for the purpose of observation, result verification, obtaining technical information or operator training.
- (h) Should the components not perform to the design or Contract specifications/requirements during any test, such failure shall be deemed a non-compliance and the Contractor shall promptly initiate the necessary corrective action. Continued failure of equipment testing to meet the requirements of the Contract specification will result in the S.O. rejecting the equipment.

G2.26.13 Documentation and Quality Assurance

- (a) Documentation shall be produced during the various stages of the commissioning, testing and handover periods. This Documentation will be a record of the quality assurance and quality control (QA/QC) aspects of the Works. The Commissioning documentation will include the following as a minimum and shall include other documentation as specified in the contract documents.
- (b) Commissioning Documentation
 - (i) Pre-Commissioning reports – Factory Acceptance Test (FAT) reports and manufacturer documentation;
 - (ii) Test Certificates;
 - (iii) Site Acceptance Test (SAT) Reports;
 - (iv) Inspection Test Plans (ITPs);
 - (v) Log sheets for recording of plant information, operating parameters, qualify parameters for presentation and interpretation of test data; and
 - (vi) Commissioning reports.
- (c) Training documentation shall include:
 - (i) Training material (including manufacturers' materials); and
 - (ii) Training survey.
- (d) Asset management documentation will include:
 - (i) CoBie data entered into the BIM model as required in G1.2 information Management;
 - (ii) Spare parts & Special tools listing and recommendations;

- (iii) Specialist Services recommendation, if any;
- (iv) As-built Drawings and BIM Models; and
- (v) O&M information.

G2.26.14 T&C Team, Roles and Responsibilities

- (a) The Contractor shall list and identify key members within the T&C Plan to be involved in the different stages of testing and commissioning. The T&C Plan shall define the proposed roles and experience of each member of the commissioning team for each stage of testing and commissioning. This team shall work with the S.O. to identify, plan, coordinate and supervise all the testing and commissioning activities for the Works. The T&C Team, as a minimum, shall include the following key personnel:
 - (i) Testing and Commissioning Manager;
 - (ii) Testing and Commissioning Engineers;
 - (iii) Mechanical, Electrical, Instrumentation and Controls Commissioning Technicians; and
 - (iv) Training Co-ordinator.
- (b) The T&C Team members shall be suitably qualified and experienced to perform the works as per the contract requirements. Qualifications of each member of the commissioning team shall be submitted to the S.O. for approval at least sixty (60) days prior to commencement of any commissioning works.
- (c) The T&C Team shall include skilled and experienced representatives of key equipment and system manufacturers during the planning, installation, commissioning and testing stages to ensure that such equipment and systems are appropriately installed and are fully tested and optimised in accordance with the manufacturers' requirements and recommendations. It is also to ensure that associated guarantees and warranties are preserved.
- (d) The T&C Team shall be supported by the respective discipline mechanical, electrical and, instrumentation and control site engineers.
- (e) Where the Contractor's employees are required by the local law or regulation to hold current licenses, certificates or authority to perform the work required of their respective positions, the Contractor shall ensure these qualifications are maintained current and valid. The Contractor shall provide continued job-related training for such employee in order to maintain such licenses, certifications or authorities to perform the work. The Contractor shall provide continued training for his employees in the areas of operation, maintenance, safety, supervisory skills and energy management. All such training shall be documented and copied to the S.O.. Should new regulations require additional licenses or certifications, the Contractor shall provide the necessary training or qualified personnel within the time frame specified in the regulations.

END OF SECTION